

Reducing Regressivity in ML-based Mass Appraisal

A practical covariance regularization layer for assessor workflows

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April 10, 2026

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Outline

- 1 Motivation
 - Regressivity in ML-based Mass Appraisal
 - Solution Approach: Regressivity Penalization
- 2 The Cook County Assessor's Office Case
- 3 Proposed Solution
- 4 Solution Implementation
- 5 Model Selection
- 6 Conclusions

Preliminaries: Mass Appraisal, Taxation & Regressivity

- ① **Mass Appraisal:** “process of estimating the value of a group of real estate properties, as of a given date, for tax purposes”

- Main source of (non-external) local revenue in the US: $\sim 30 - 50\%$
- Affects almost everyone: directly or indirectly (public services, transport, rent, etc.)

- ② **Taxation Mechanism:**

- **Assessment:** Local assessors estimate “market value” of properties
- **Tax Base Adjustment:** Apply homestead exemptions (e.g., primary residence)
- **Tax Rate:** Set by local governments (city, county, etc.)
 - E.g., Cambridge: $0.55 - 0.6\%$, Boston: $1 - 1.1\%$, Chicago: $2 - 2.3\%$.
- **Tax Calculation:** $\text{Property Tax} = (\text{Assessed Market Value} - \text{Exemptions}) \times \text{Tax Rate}$

- ③ **ML-based Assessment:** Assessors offices are moving from linear regression-based methods to more “advanced” ML models.

- Tree ensembles.
- Neural Networks.

Preliminaries: Mass Appraisal, Taxation & Regressivity

- ➊ **Assessors Concerns with ML:** model explainability, operational feasibility, and **fairness/equity**.
 - “Black-box” models are hard to interpret/defend
 - Costly to maintain and deploy
 - Can enforce unfair behaviors

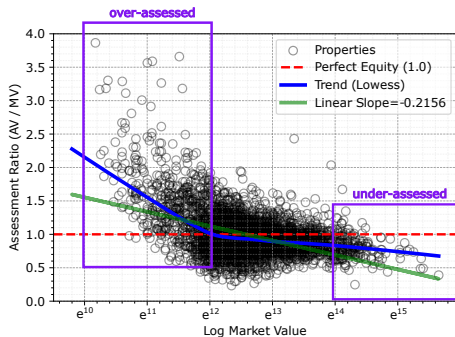
- ➋ **Fairness/equity** is a priority & has been well-studied (even before ML).
Assessments should be *fair* across properties of:
 - Similar characteristics (*Horizontal Equity*).
 - Different price levels (*Vertical Equity*).

- ➌ **Regressivity:** Lower-tier properties tend to be over-assessed (assessment > market price).
Higher-tier properties tend to be under-assessed (assessment < market price).
 - *Regressivity* is very common in real-world assessments
 - *Progressivity* is the opposite, but is less common

Regressivity in ML-based Mass Appraisal

How does regressivity look like?

Regressivity in LightGBM



Notation:

- AV = Assessed Values
- MV = Market Values
- $r = \frac{AV}{MV}$ = Assessment Ratios.

Trend Dependency $\rightarrow$ Regressivity

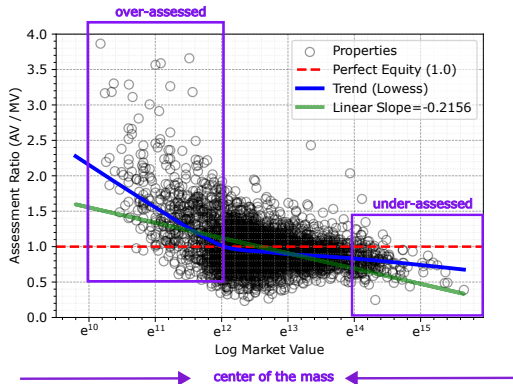
If trend is negative (MV vs r),

$\Rightarrow$ we have a **regressive behavior**.

*Note: If **positive** we have progressive behavior.*

Regressivity in ML-based Mass Appraisal

Regressivity is the expected behavior in regression



Note: The Bayes predictor $f(X) = \mathbb{E}[y|X]$ satisfies, $\text{Cov}(e, Y) = -\mathbb{E}[\text{Var}(Y|X)] \leq 0$, where $e = f(X) - Y$.

Solution Approach: Regressivity Penalization

- A **Multi-objective problem**.
 - **Prediction Error**: Estimate *market values* as precise as possible.
 - **Fairness**: Avoid systematic bias across property values (regressivity).
- What can we do to correct the regressive behavior?
 - Constraint/penalize negative dependence between r (assessment ratio) and y (sale price): .

Old Paradigm: Minimize in-sample prediction error

Let y be the price objective, X a property-related feature matrix and $\mathcal{H}$ model family class. Solve

$$f_{\text{base}} \in \arg \min_{f \in \mathcal{H}} \{L(f, X, y)\}, \quad (1)$$

Regressivity-Aware Paradigm: Add dependence penalization

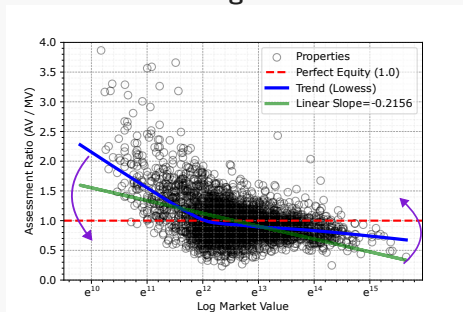
Let $r = f(x)/y$ be the assessment ratios, and Ψ be some dependence measure between vectors. Solve

$$f_{\text{dep}} \in \arg \min_{f \in \mathcal{H}} \{L(f, X, y) + \rho \Psi(r(f), y)\}, \quad (2)$$

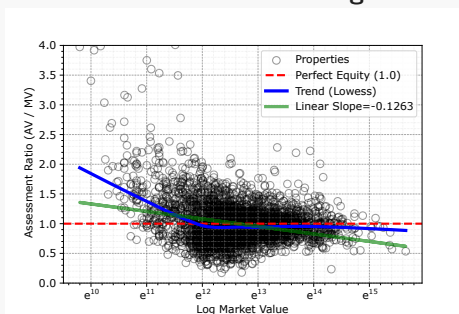
Solution Approach: Regressivity Penalization

Penalized Objective: $\text{Error} + \rho \cdot \text{Covariance}(\text{ratios}, \text{prices})^2$

Baseline LightGBM



Covariance-Penalized LightGBM

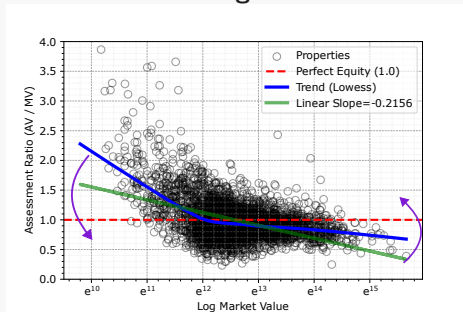


Corr(ratios, prices):	-0.2666	→	-0.1224
slope(ratios~log-prices):	-0.2156	→	-0.1263

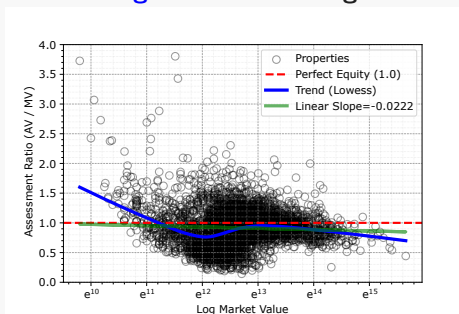
Solution Approach: Regressivity Penalization

Penalized Objective: $\text{Error} + \rho \cdot \text{Surrogate}(\text{ratios}, \text{prices})$

Baseline LightGBM



Surrogate-Penalized LightGBM



Corr(ratios, prices):	-0.2666	→	-0.0239
slope(ratios~log-prices):	-0.2156	→	-0.0222

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 - Past models: linear vs nonlinear model
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Cook County Case & Data: Assessment Year 2025

- **The Cook County Assessor's Office** manages an extensive repository ([model-res-avm](#)).
 - We mimic their pipeline; available in [assessment-year-2025](#) (dataset, preprocessing, feature engineering, etc.)
- **Data:** consists of $> 350K$ observations and ~ 200 raw features: property features, public services access, distance to parks, transport accessibility, etc.
- **Data splits:** we use 2024 as the "current year" (what we want to assess), 2023 as the test set, and the previous years (2016-2022) for training/validation.

Table: Data splits performed on the 2025 dataset.

Split	Begin Date	End Date
Train/Validation	Jan. 01, 2016	Dec. 31, 2022
Test	Jan. 01, 2023	Dec. 31, 2023
Assessment	Jan. 01, 2024	Dec. 27, 2024

-  **Note:** There may still be mismatches between our code and the one available in the source [assessment-year-2025](#), in terms of functions, objects, data-points, dates, general code, etc.

Past models: linear vs nonlinear model

Baseline Models Performance:

Model	R^2	MAE	COD	COV (IAAO)	VEI	PRD	PRB	MKI
Linear Regression	0.798	\$85,273	24.306	0.354	6.223%	1.016	0.033	1.009
LGBM Regressor	0.860	\$73,398	21.647	0.330	-31.431%	1.082	-0.108	0.890

Table: Test set metrics: Accuracy (R^2 , MAE), horizontal equity (COD, COV), and vertical equity (VEI, PRD, PRB, MKI).

Reference Values (Vertical Equity)

VEI	PRD	PRB	MKI
$[-10\%, 10\%]$	$[0.98, 1.03]$	$[-0.05, 0.05]$	≈ 1

Table: IAAO recommended for VEI, PRD, and PRB.



Intuition

- Linear models: **stable slope** and **interpretable**, but **misses nonlinear market structures**.
- Nonlinear models: stronger **accuracy + dispersion**, but can **worsen vertical equity** unless controlled.

Past models: linear vs nonlinear model

Can we achieve a middle-ground solution?

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3 Proposed Solution

- Middle-ground solution: Covariance Penalization Step
- Performance review: Covariance-Penalized LightGBM
- Tradeoffs: Vertical Equity (PRD & VEI) vs Accuracy
- Performance Conclusion: Covariance-Penalized LGBM

4 Solution Implementation

5 Model Selection

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Proposed Solution: Regressivity Correction

- ❶ A **regressivity-aware penalization**, that reduces regressivity, while maintaining similar accuracy & dispersion level (sometimes even improving them).
 - Can be used in many regression models (linear regression, boosting, NN, etc.)
 - Maintains the same principles of the original model, training process, interpretability, etc.

- ❷ A solution that can be implemented with **minimal changes to current pipelines**
 - Add only 1 extra hyperparameter.
 - Can be used during or after calibration of the original model.

- ❸ An **alternative model selection step**, that can secure better tradeoffs between accuracy/dispersion & regressivity constraints.
 - Easier to interpret: can enforce explicit constraints (e.g., $PRD \in [0.98, 1.03]$).
 - Is a more flexible modeling process.

Middle-ground solution: Covariance Penalization Step



Solution Approach

1 Take $\hat{y} = f(X)$, an accurate model (low error & dispersion).

2 Add a covariance regularization term:

$$\text{Predictive Error} + \rho \cdot \text{Cov}(\text{ratios}, \text{price})^2$$

3 Train the new model, minimizing the penalized objective with $\rho > 0$.

4 Select (tune) ρ that achieves a middle-ground solution in validation sets.

**Technical Note:* This penalization can be “added” to any model that aim for $f^*(X) := \arg \min_{f \in \mathcal{H}} \mathcal{L}(f; X, y)$ (e.g., minimize MSE). We can just re-write it as $f_{cov}^*(X) := \min_{f \in \mathcal{H}} \mathcal{L}(f; X, y) + \rho (\text{Cov}(f(x_i)/y_i, y_i))^2$.

Middle-ground solution: Covariance Penalization Step

Penalization Definitions

1 Empirical Covariance Penalization:

$$\widehat{\text{Cov}}(r, y)^2 = \left(\frac{1}{n} \sum_{i=1}^n (r_i - \bar{r})(y_i - \bar{y}) \right)^2$$

- It explicitly controls linear dependencies between r and y .
-  Aggregation of positive-negative trends can hide correlations.
-  It is a coupling term (non-separable).

2 Surrogate Penalization:

$$\text{Surr}(r, y) = \frac{1}{n} \sum_{i=1}^n (r_i - 1)^2 (y_i - \bar{y})^2$$

- Its an approximation of an upper bound over the covariance.
-  Controls any deviation of r from 1 (weighted by the distance to the average price $\bar{y}$).
-  It is separable!

**Note: For simplicity, we refer to both as "Covariance" penalization, since both are aiming to reduce this quantity.*

Performance review: Covariance-Penalized LightGBM

Baseline Objective: Prediction Error

Model	Penalty ρ	R2	MAE	COD	COV (IAAO)	VEI	PRD	PRB	MKI
LinearRegression	-	0.798	\$85,273	24.306	0.354	6.223%	1.016	0.033	1.009
LGBMRegressor	-	0.860	\$73,398	21.647	0.330	-31.431%	1.082	-0.108	0.890

Table: Test results for Accuracy (R2, MAE), horizontal equity (COD, COV), and vertical equity (VEI, PRD, PRB, MKI).

Penalized Objective: Pred. Error + $\rho \cdot \text{Covariance}(\text{ratios, prices})^2$

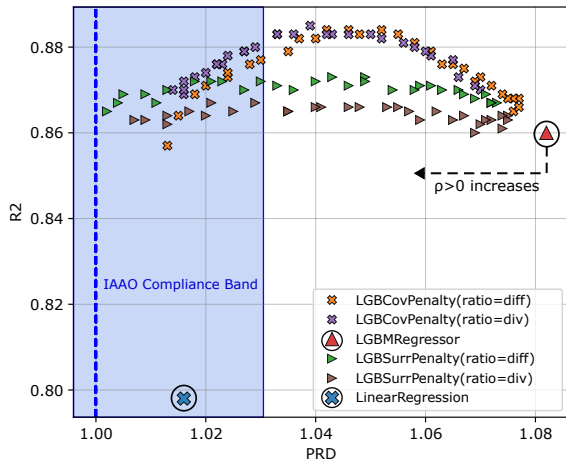
Model	Penalty ρ	R2	MAE	COD	COV (IAAO)	VEI	PRD	PRB	MKI
LGBMSurrPenalty [v1]	2.043	0.872 (+1.40%)	\$74,579 (-1.61%)	21.770 (-0.57%)	0.316 (+4.24%)	-0.043% (+99.86%)	1.030 (+63.41%)	+0.003 (+97.22%)	0.970 (+72.73%)
LGBMSurrPenalty [v2]	280.7	0.867 (+0.81%)	\$75,641 (-3.06%)	21.829 (-0.84%)	0.313 (+5.15%)	+3.403% (+89.17%)	1.029 (+64.63%)	+0.008 (+92.59%)	0.970 (+72.73%)
LGBMCovPenalty [v1]	6.58	0.877 (+1.98%)	\$72,042 (+1.85%)	21.693 (-0.21%)	0.319 (+3.33%)	-6.579% (+79.07%)	1.030 (+63.41%)	-0.026 (+75.93%)	0.989 (+90.00%)
LGBMCovPenalty [v2]	1487.0	0.880 (+2.33%)	\$70,855 (+3.46%)	21.467 (+0.83%)	0.317 (+3.94%)	-4.043% (+87.14%)	1.029 (+64.63%)	-0.022 (+79.63%)	0.986 (+87.27%)

Table: Test behavior. Percentual change vs LGBMRegressor in parentheses (**improvement** / **worsening**). For MAE/COD/COV: lower is better. For VEI/PRB: closer to 0 (reduction in $|x|$). For PRD/MKI: closer to 1 (reduction in $|x - 1|$).

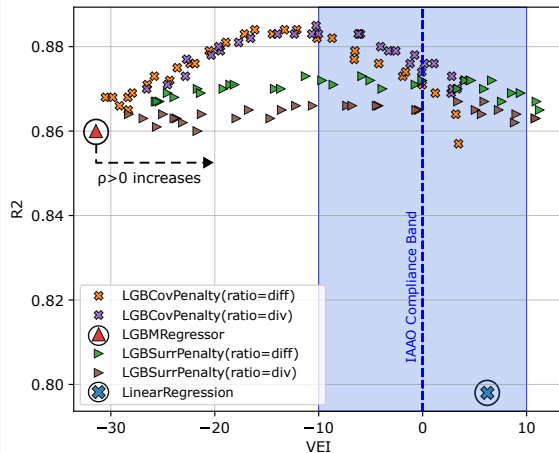
Tradeoffs: Vertical Equity (PRD & VEI) vs Accuracy

Pareto Frontier: Test set behavior for different penalization terms ($\rho > 0$)

Trade-off: Vertical Equity vs Accuracy (PRD vs R2)



Trade-off: Vertical Equity vs Accuracy (VEI vs R2)



Can we achieve a middle-ground solution? → Yes.



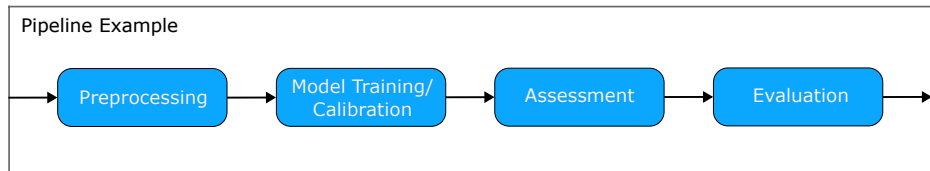
Performance Conclusion: A middle-ground solution is feasible

- Can be competitive in **accuracy** and **horizontal equity**. ✓
- Can be competitive in **vertical equity**. ✓
- Can improve metrics at least up to IAAO's **compliance ranges**. ✓

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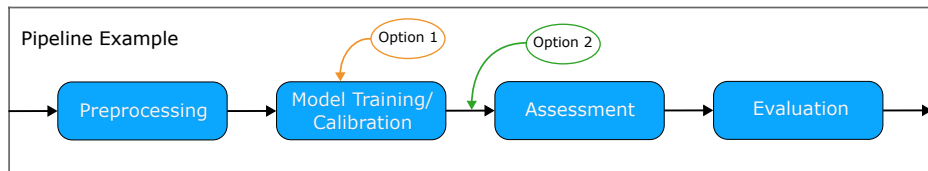
Is this solution easy-to-deploy in current settings?



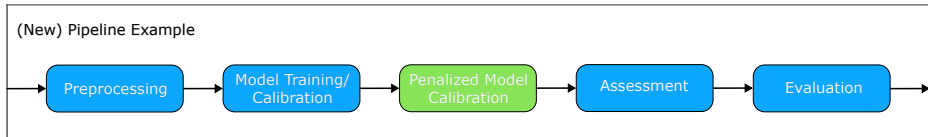
Implementation Complexity

The solution can be implemented in **two different ways** in current workflows:

- ➊ **Option 1:** Calibrate penalization ρ along with the rest of the model's parameters.
 - ✓ Straightforward training process.
 - ⚠ Modifies the current selection criteria (accuracy-based) to tackle different objectives (accuracy/dispersion/regressivity).
- ➋ **Option 2:** Calibrate penalization ρ by itself, after selecting the best-accuracy model f^* .
 - ✓ Implementation is less invasive (easy-to-deploy). → same selection criterion.
 - ✓ Only 1 hyperparameter to tune independently.
 - ⚠ It can be sub-optimal in tradeoff/Pareto-sense (*Note: still good in practice*).



Is this solution easy-to-deploy in current settings? → Yes.



Implementation Conclusion: It can be implemented with minimal changes

- Usable in the **same existing pipelines**, as an additional step or hyperparameter tuning. ✓
- Only **adds 1 extra parameter** with **clear behavior patterns**. ✓
- Can be **calibrated/selected after model tuning** (independent calibration). ✓

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 - Model Selection: Key challenges
 - Model Selection: A Pareto-optimal approach
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Can we just select the best performing model in average? → It depends.

Model Selection: Key challenges

Tackling a **multi-objective problem**

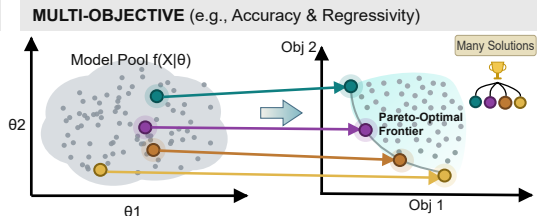
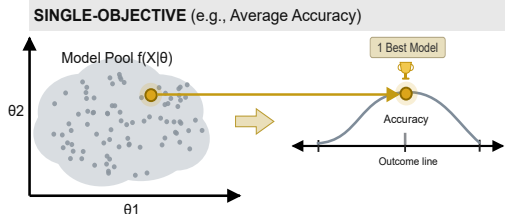
- 1 We aim to control multiple objectives:

Accuracy $\leftrightarrow$ dispersion (horizontal equity) $\leftrightarrow$ regressivity (vertical equity).

- 2 What is the “best average performance” in this case? $\rightarrow$ Trade-off dependent.

- 3 There are many ways to define the “best” model:

- Old method: Select the best accuracy in average (⚠ misses the rest of the metrics).
- Pareto-optimal: Filter the models that do not satisfy compliance ranges for regressivity (PRD, VEI, etc.), and select the best accuracy in average (✅ a more natural choice).



Model Selection: A Pareto-optimal approach

Multi-objective regressivity-constrained model selection

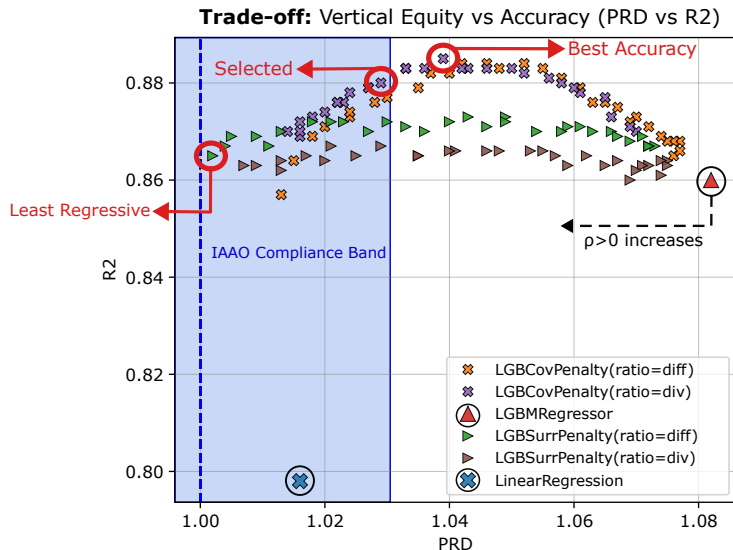
- 1 **Train:** $c \in \mathcal{C}$ configurations of the model.
 - E.g.: Different models, hyperparameters, penalty $\rho > 0$.
- 2 **Evaluate:** on $v \in \mathcal{V}$ validation folds, some
 - Prediction error metric $\mathcal{E}_v(c)$ (e.g., MSE), and some
 - Regressivity metric $\mathcal{F}_v(c)$ (e.g., PRD).
- 3 **Calibrate:** Solve the sorting problem (2-objectives example)

$$\begin{aligned} c^* &:= \arg \min_{c \in \mathcal{C}} \quad \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \mathcal{E}_v(c) \\ \text{s.t.} \quad &\mathcal{F}_v(c) \leq \tau, \quad \forall v \in \mathcal{V}, \end{aligned}$$

or some other aggregation of the results $\mathcal{F}_v(c)$, for $v \in \mathcal{V}$.

- Note:*
- (1) c^* is a Pareto-optimal solution, in the validation folds in $\mathcal{V}$.
 - (2) We can add as many equity metrics as needed as constraints (e.g., COD, PRB, VEI, etc.)
 - (3) If we allow a stacking of models $f(X) = \sum_{c \in \mathcal{C}} w_c f_c(X)$, the trade-offs can be further improve.

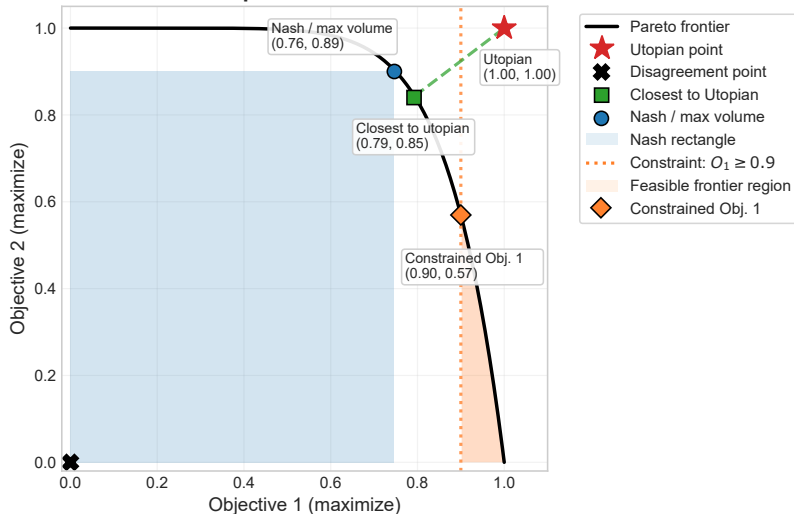
Model Selection: Different Tradeoff Solutions



- The constrained model-selection is **not perfect**.
 - It does not secure constraint satisfaction in future out-of-sample sets.
 - The constraints may not be feasible in the validation windows.
 - One may sacrifice too much accuracy to satisfy the constraints.
- **Alternative criteria:**
 - Closest-to-Utopian:
 - Define a normalized ideal point (best for all metrics), and minimize distance to it.
 - Nash Equilibrium:
 - Maximize the volume of a hyper-cube inside the Pareto-efficient frontier.

Model Selection Alternatives

Pareto Frontier Example: Tradeoffs & Solution Alternatives



Model Selection Results

Table: Held-out test set performance. The penalty parameter ρ is specific to the penalty definition.

Solution	Selected model	ρ	R^2	MAE	Med. r	Mean r	COD	COV	PRD	PRB	VEI
Unconstrained baselines											
BASLINE	Linear Reg.	—	0.762	\$86,244	1.001	1.050	25.01%	36.86%	1.039	-0.011	-13.53%
BASLINE	LGBM	—	0.885	\$71,912	0.944	1.021	23.07%	35.48%	1.085	-0.118	-37.64%
Constraints: PRD											
CONSTRAINED	LGBSmooth	5.96	0.881	\$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%
UTOPIA	LGBSmooth	100.00	0.877	\$79,217	0.930	0.953	24.47%	34.89%	1.019	0.032	-4.41%
NASH	LGBSmooth	2.95	0.883	\$73,395	0.938	0.988	22.63%	33.63%	1.048	-0.034	-12.20%
Constraints: PRD, PRB, and VEI											
CONSTRAINED	LGBSmooth	5.96	0.881	\$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%
UTOPIA	LGBSmooth	7.91	0.882	\$74,939	0.935	0.974	23.02%	33.85%	1.036	-0.007	-8.36%
NASH	LGBSmooth	5.96	0.881	\$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%
Constraints: PRD, Mean r, and COD											
INFEASIBLE*	—	—	—	—	—	—	—	—	—	—	—
UTOPIA	LGBSmooth	6.87	0.882	\$74,449	0.937	0.977	22.84%	33.72%	1.038	-0.011	-9.38%
NASH	LGBSmooth	2.95	0.883	\$73,395	0.938	0.988	22.63%	33.63%	1.048	-0.034	-12.20%
Constraints: Full metric set[†]											
INFEASIBLE*	—	—	—	—	—	—	—	—	—	—	—
UTOPIA	LGBSmooth	5.96	0.881	\$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%
NASH	LGBSmooth	5.96	0.881	\$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%

*INFEASIBLE abbreviates CONSTRAINED INFEASIBLE; no feasible selected solution.

[†] PRD, PRB, VEI, COD, COV, Mean r , Median r , and Weighted Mean r .

Could we constraint the learning model directly instead?

Constrained ML: three main challenges

- 1 **Feasibility / learnability (Train):** Hard constraints like “ $PRB = 0$ ” (perfectly flat trend) can be infeasible or unstable in-sample (especially for simpler models like Linear Regression).
- 2 **Generalization / robustness (Test):** Constraints tied to complex dependencies can fail in out-of-sample sets (selection bias + small population sample of “market values”).
- 3 **Implementation:** Hard constraints that are non-convex/non-differentiable (e.g., VEI) require a much more sophisticated machinery in terms of implementation.

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 - Appendix: Metrics Definitions (Accuracy, horizontal & vertical equity)

Conclusions: Review

Main motivation

High accuracy does not guarantee vertical equity → we can enforce a practical notion of regressivity-control, in addition to accuracy, during training.

Proposed Solution

A **plug-in vertical-equity layer** for general regression learners:

$$\text{loss} + \rho \cdot \Psi(r, y)^2,$$

to reduce price-tier inequities (regressivity) in ML-based Mass Appraisal.

Why it fits assessor's workflows?

- Works with currently used models (e.g., Linear Regression & Gradient Boosting Machines).
- One knob $\rho > 0$ & familiar diagnostics (ratio plot, PRD/PRB, COD).
- Provides a clear tradeoff curve for policy/operational choice.

Key Limitations of the Proposed Solution

- 1 The method **does not constraint the regressivity measures** directly (PRD, PRB, VEI, MKI, etc.)
- 2 We currently have **no guarantees over the regressivity-control** in out-of-sample sets.
- 3 In principle, a covariance penalty only aims at **linear relationships** between r and y ; not more complex relationships (we could be enforcing them).
- 4 The penalty term is **unaware of sub-market and geographical structures**, that could be enforcing a local regressivity behavior.

Conclusions: Extensions

Beyond linear dependence

- Penalize **nonlinear dependence measures** (kernels, HSIC/maximal correlation/dCov, etc.).
- Move from log-price models to **price-scale models penalization**: different risk profiles and interpretability.

⚠ Challenges: Training/learnability & generalization.

Data stratification awareness

- **Submarket-aware penalties** (stratified constraints), to avoid averaging away local inequities.
- **Sensitive attribute control** (neighborhood, class, etc.) on top of price-tier fairness.

⚠ Challenges: How to choose them in a fair & unbiased way?

Interpretability

- Interpretable learners (EBM/GAM) with vertical-equity control, or post-training explanation schemes for nonlinear learners (LGBM, NN, etc.)

⚠ Challenges: A completely new research direction.

Questions?

Happy to discuss: the implementation in ML objectives, extensions, metric reporting, etc.

Appendix: Metrics Definitions (Accuracy, horizontal & vertical equity)

Assessment ratios statistics:

$$r_i := \frac{AV_i}{MV_i} = \frac{\widehat{MV}_i}{MV_i}, \quad \tilde{r} := \text{median}(r), \quad \bar{r} := \frac{1}{n} \sum_{i=1}^n r_i, \quad \bar{r}_w = \frac{\sum_i MV_i r_i}{\sum_i MV_i} = \frac{\sum_i AV_i}{\sum_i MV_i}$$

Accuracy

$$R^2 = 1 - \frac{\sum_i (MV_i - \widehat{MV}_i)^2}{\sum_i (MV_i - \overline{MV_i})^2},$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |MV_i - \widehat{MV}_i|$$

Horizontal Equity / Dispersion (IAAO)

$$\text{COD} = 100 \cdot \frac{\frac{1}{n} \sum_i |r_i - \tilde{r}|}{\tilde{r}},$$

$$\text{COV} = \frac{\text{SD}(r)}{\bar{r}}$$

Vertical Equity (IAAO & CCAO)

$$\text{VEI} \approx 100 \cdot \frac{\text{median}(r)_{\text{last decile}} - \text{median}(r)_{\text{first decile}}}{\tilde{r}}$$

(Interpretation: $\text{VEI} < 0$ regressive, $\text{VEI} > 0$ progressive)

$$\text{PRD} = \frac{\bar{r}}{\bar{r}_w},$$

$$\text{PRB} = \text{slope} \left(\frac{r_i - \tilde{r}}{\tilde{r}} \sim \log_2 \left(\frac{1}{2} (MV_i + AV_i / \tilde{r}) \right) \right)$$

$$\text{MKI} = \frac{\text{Gini}(AV)}{\text{Gini}(MV)}$$

Appendix: General-learner with covariance constraint / penalty

Base learning problem (the model already being used)

Given hypothesis class $\mathcal{H}$ (Linear Regression, Gradient Boosting, Neural Networks, etc.), regression models solve

$$f^*(X) = \arg \min_{f \in \mathcal{H}} \mathcal{L}(f; X, y) = \arg \min_{f \in \mathcal{H}} J(f; X, y) + \Omega(f),$$

with J empirical risk (typically MSE), Ω complexity control (regularizer), and $f^*(X) = \hat{y}$ the estimator.

Add a vertical-equity dependence control

Let $g(\hat{y}, y)$ be some assessment ratio proxy and $\phi(y)$ a value transform of the market value. We solve

$$\min_{f \in \mathcal{H}} \mathcal{L}(f; X, y) \quad \text{s.t.} \quad (\widehat{\text{Cov}}(g(\hat{y}, y), \phi(y)))^2 \leq \varepsilon,$$

or the soft penalty version:

$$\min_{f \in \mathcal{H}} \mathcal{L}(f; X, y) + \rho(\widehat{\text{Cov}}(g(\hat{y}, y), \phi(y)))^2.$$

Practical meaning: take an existing assessor model, add one tunable term that discourages price-tier bias (linear trend).

Appendix: choosing g, ϕ and dealing with coupling

Choosing $g(\hat{y}, y)$ and $\phi(y)$

Common training is in log space: $y = \log(\text{MV})$, $\hat{y} = f(X)$.

- Log-space proxies:

$$g(\hat{y}, y) = \hat{y} - y \quad (\text{log-ratio, target 0}),$$

or a log-ratio proxy:

$$r = \hat{y}/y, \quad g(\hat{y}, y) = r \quad (\text{target 1}).$$

- We start with $\phi(y) = y$ (log market value); extend in future work.

Coupling of the covariance term

$$\widehat{\text{Cov}}(g, \phi) = \frac{1}{n} \sum_{i=1}^n (g_i - \bar{g})(\phi_i - \bar{\phi})$$

- $\widehat{\text{Cov}}(g, \phi)$ is **global** $\Rightarrow$ non-separable.
- Exact Hessian is dense; many libraries expect per-sample Hessians.

Separable surrogate (upper bound)

Using a Cauchy-Schwarz/Jensen's inequality:

$$\widehat{\text{Cov}}(g, \phi)^2 \leq \frac{1}{n} \sum_{i=1}^n (g_i - \bar{g})^2 (\phi_i - \bar{\phi})^2.$$

Appendix: Correlation Metrics Evolution

Distributional Correlation Metrics Evolution vs rho (Test)

